RIZVI COLLEGE OF ENGINEERING

Department of Artificial Intelligence & Data Science

Assignment No. 02

Name	Sakpal Girish Dayaghan	UIN	231A011
Class	B.E (AI&DS), Semester VII	Date of submission	25 October 2026

Q.1 "The bank will not open until the river recedes."

(a) Type of lexical ambiguity

The word that causes the trouble is **bank**. This is **homonymy**: the spelling is the same but the meanings are unrelated, like a bank that keeps money and the bank of a river. (If the meanings were related, such as the bank as an organisation and the bank as a building, it would be polysemy instead.) Since we have to choose between different senses of one word, this is lexical ambiguity, not structural ambiguity. Knowing that bank is a noun here helps a little, because it rules out verb meanings like "to bank a plane", but several noun senses are still left.

What makes this sentence a good example is that its clues disagree. *River* and *recedes* pull toward the riverbank, but *will not open* pulls toward a place of business. I think the sensible reading is the **financial bank**: the branch is near the river, there is a flood, and it cannot open until the water goes down. So "river" is a misleading clue.

(b) How WordNet or BabelNet could help

WordNet stores each meaning of a word as a synset, with a short definition and links to related synsets. "Bank" has around ten noun senses, and the three that matter here are the sloping land beside a body of water, the institution that accepts deposits, and the building where banking is done. This is how I would pick the right one.

1. Tag "bank" as a noun, which removes the verb senses.
2. Take the other content words, *open*, *river* and *recede*, and compare them with the definition of each noun sense and with the definitions of related synsets such as the hypernyms. This is the extended Lesk method. "River" would match the riverbank sense through "body of water".
3. Add a rule about the verb "open". Things that open for business are shops, offices and institutions, so this supports the institution and building senses.
4. Add up the evidence and choose the sense with the best total score.

BabelNet works in the same way but is bigger. It joins WordNet with Wikipedia and other resources into multilingual synsets, so a concept is linked to its Hindi or Marathi word too, and it has many more related concepts and named entities. Links between articles on floods, rivers and banks could give the extra evidence that connects "river recedes" to a bank building.

(c) Why automatic systems can still fail

Even with these resources a system can go wrong here. A plain overlap count would probably choose the riverbank, because "river" matches its definition directly, and that would be the wrong answer.

The real reason the bank cannot open, that floods close buildings near rivers, is common sense and is not written in any definition. The sentence is also short, so there is very little evidence to score. WordNet splits meanings very finely, and new or specialised uses may be missing altogether. "The bank" can also mean the building, the staff or the company all at once (metonymy), which a list of senses cannot capture. Finally, Lesk-style methods need the same words to appear, so a paraphrase like "the water level goes down" would be missed.

Q.2 Word Sense Disambiguation with Naïve Bayes

(a) Applying Naïve Bayes to WSD

Naïve Bayes treats WSD as a classification problem, with one classifier for each ambiguous word. The classes are the senses of the word, and the features are the words around it, usually inside a window of a few words on each side. Sometimes the part of speech of the neighbours is added as well. For a given context, the classifier picks the sense that gives the highest value of:

$$P(s) \times P(f1 \mid s) \times P(f2 \mid s) \times \ldots \times P(fn \mid s)$$

Here s is a sense and f1 to fn are the context features. Both kinds of probability come from counting in a sense-tagged corpus such as SemCor. I would use add-one smoothing so that an unseen word does not make everything zero, and add log probabilities instead of multiplying, to avoid numbers becoming too small. Below is a tiny example for the context words *river* and *flood*. The numbers are made up, only to show the working.

Sense of "bank"	P(s)	P(river | s)	P(flood | s)	Score
Financial institution	0.70	0.01	0.02	0.70 x 0.01 x 0.02 = 0.00014
Riverside land	0.30	0.20	0.10	0.30 x 0.20 x 0.10 = 0.006

Table 1: Example Naïve Bayes scores (invented numbers)

The riverside sense wins easily. This is the same trap as in Q.1: words that strongly go with one sense can outvote the correct one.

(b) Limitations compared with Lesk's knowledge-based approach

Point	Naïve Bayes	Lesk
Data needed	Needs many sense-labelled examples for every word, which is expensive to make	Needs only a dictionary or WordNet definitions, no labelled corpus
Coverage	Only works for words and senses seen in training, and rare senses get few examples	Works for any word that has a definition
Assumption	Treats context words as independent given the sense, and ignores word order	No independence assumption, but also ignores syntax
Bias	The prior favours the most frequent sense, and the model is tied to its training domain	Not tied to a domain, but depends on how good the definitions are
Scalability	A separate model for every target word	One method for all words
Accuracy	Usually better when enough labelled data exists	Usually lower, because definitions are short and overlap is rare

Table 2: Naïve Bayes compared with Lesk

In short, if there is enough labelled data Naïve Bayes is normally more accurate. Its main weakness against Lesk is that it depends on that data and copes badly with rare or unseen senses, while Lesk needs no labelled data at all.

(c) Improvement using a semi-supervised method (Yarowsky's algorithm)

The company can cut its labelling cost by using Yarowsky's bootstrapping idea. It rests on two observations. A strong collocation such as "bank account" almost always has just one sense, and a word usually keeps one sense inside a single document.

I would start with a few seed examples for each word, for instance "bank account" for the institution and "river bank" for the riverside sense, and train the Naïve Bayes classifier on them. Then I would run it on a large unlabelled corpus and keep only the predictions with high confidence, say above 0.9. These get added to the training set and the classifier is retrained. This repeats until no new examples pass the threshold. At the end, the one-sense-per-discourse idea can fix any label that disagrees with the dominant sense in the same document.

The gain is that a handful of labelled examples plus plenty of ordinary text can come close to a fully labelled set. The danger is error drift, where a wrong label gets reinforced in every round. A strict threshold, keeping the seed labels fixed, and checking on a small held-out labelled set all help to limit it.

Q.3 "John met Alex after he finished the project."

(a) The anaphora and why resolution is hard

The anaphor is the pronoun **"he"**, and the possible antecedents are **John** and **Alex**. It is a pronominal anaphora, and since the antecedent is in the same sentence it is also intrasentential.

The reason it is hard is that the usual filters cannot separate the two candidates. Both names are singular and masculine, so gender and number agreement keep both. Both readings are also grammatical, because the pronoun sits in a subordinate clause and is allowed to refer to the subject of the main clause. After that, the answer depends on what actually happened. John may have finished the project and then met Alex, or Alex may have finished it and John met him afterwards to review it. Nothing in the sentence settles this, so a person would look at the earlier discourse or use knowledge of the situation. Preferences like "the subject is more likely" give a guess, not a certain answer.

(b) How Hobbs' algorithm would resolve "he"

Hobbs' algorithm works only on the syntax tree. It searches the tree in a fixed order and proposes the first noun phrase that agrees with the pronoun. For this sentence I assume the clause "after he finished the project" attaches to the verb phrase, so the tree is:

[S1 [NP John] [VP met [NP Alex] [SBAR after [S2 [NP he] [VP finished [NP the project]]]]]]

1. Start at the NP that dominates "he" and go up to the first NP or S node. That is S2, so X = S2.
2. Search the branches of S2 to the left of the path. There is nothing to the left of "he", so no candidate is found.
3. S2 is not the highest S node, so go up to the next NP or S node. The path passes through SBAR and VP and reaches S1, which becomes the new X.

4. Search the branches under S1 to the left of the path, left to right and breadth first. The first NP found is **John**. Alex is inside the VP, which is on the path, so it is not reached first.
5. Check agreement. John is singular and masculine, like "he", so he is accepted.

So Hobbs' algorithm resolves "he" to **John**, and it would do so every time for this tree. It never looks at meaning, so if the real answer is Alex it is simply wrong. The result also depends on a correct parse. If "after he finished" were attached higher in the tree, the search order could change.

(c) A real-world scenario where syntax alone does not give coherence

Think of a line in a tenancy agreement or a customer email: *"The tenant told the landlord that he would repair the roof before the monsoon."* By syntax alone, "he" can be the tenant or the landlord. Both are grammatical and both agree in gender and number. Which one is meant depends on who is normally responsible for roof repairs, on the text before it, and on what happened next. If a contract analysis tool picks the wrong person, it can give the obligation to the wrong party, and that could lead to a real dispute.

The well-known Winograd example shows the same thing. In "The council refused the demonstrators a permit because they feared violence", changing "feared" to "advocated" switches who "they" refers to, with no change in syntax. I also saw a small case of this in my own Assignment 1 tests. In Test 4 I wrote "He was alone", meaning the brother from the previous turn, and the ChatGPT NPC did not acknowledge it. Reference across turns needs discourse and world knowledge, not only syntactic rules.